PREDICTIVE MODELING OF SCHOOL DROPOUT IN PAKISTAN

Midterm Group Project | Machine Learning for Public Policy
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Data Source: UNICEF MICS Round 6 (2017-2020)
Provinces: Punjab · Sindh · Khyber Pakhtunkhwa · Balochistan
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EXECUTIVE SUMMARY

Pakistan faces one of the world's most severe school dropout crises. This study applies a hybrid machine learning and explainable AI framework to predict school dropout risk using UNICEF MICS Round 6 (2017-2020) data from four provinces: Punjab, Sindh, Khyber Pakhtunkhwa (KPK), and Balochistan. After preprocessing 810,026 household member records and applying education-level capping, age filtering, and deduplication, 131,748 school-age children (ages 6-24) were retained for analysis. The national dropout rate across these provinces stands at 23.6%, substantially higher than the 18.7% documented in Bangladesh by Khatun et al. (2025).

XGBoost emerged as the superior predictive model with 86.7% accuracy and a ROC-AUC of 0.911 across 10-fold cross-validation, outperforming both the logistic regression baseline (AUC=0.886) and Random Forest (AUC=0.853). An original engineered feature, the Age-Grade Mismatch Index (AGM), emerged as the single most important predictor across all methods, with students severely behind their expected grade facing dropout rates 9.1x higher than on-track peers (44.5% vs 4.9%). Provincial disparities are stark: Balochistan records 32.1% dropout versus KPK at 21.0%, and the province-stratified model AUC gap (Punjab 0.928 vs Balochistan 0.841) quantifies the structural supply-side barriers in Balochistan that household survey data cannot capture.

DATA DICTIONARY

The following table maps MICS SPSS variable codes to their analytical counterparts and the encoding applied in this study.

MICS Code	Variable Name	Description	Encoding
HL6	Age	Student age in years	Numeric (6-24)
HL4	Sex	Sex of household member	Sex_MALE: 1=Male, 0=Female
ED9	Attend_school_thisYear	TARGET: attended school this year	1=Attending (76.4%), 0=Dropout (23.6%)
ED6	Completed_grade	Whether student completed current grade	1=YES, 0=NO (recoded from 1/2)
ED16B	Last_education_grade	Highest grade attended (previous year)	Numeric grade 1-18+ (replaced by AGM)
HH6	Area	Urban or rural locality	Area_URBAN: 1=Urban, 0=Rural
division	Division	Administrative region within province	One-hot encoded (drop first)
windex5	Wealth_index	Household wealth quintile	0=Poorest to 4=Richest (recoded)
HH15	Language	Interview language (ethnicity proxy)	Language_other: 1=Non-majority, 0=Majority
melevel	Mother_s_education	Mother's highest education level	0=None to 4=Higher (label encoded, capped)
felevel	Father_s_education	Father's highest education level	0=None to 4=Higher (label encoded, capped)
derived	Age_Grade_Mismatch	Engineered: Age - (Last_grade + 6)	Continuous [-18, 17]; positive = behind expected grade
Province	Province	Administrative province (4 levels)	One-hot (Balochistan=reference)

1. INTRODUCTION

1.1 Policy Context and Motivation

Pakistan has one of the largest out-of-school child populations globally. The MICS Round 6 data analyzed in this study reveals a national school dropout rate of 23.6% among children aged 6-24 across four provinces, a figure that substantially exceeds Bangladesh's 18.7% and reflects Pakistan's comparatively lower education infrastructure, higher rates of child labor, and deep gender and geographic disparities in access to schooling.

Provincial governments and the federal Ministry of Federal Education and Professional Training implement stipend programs and conditional cash transfers through the Benazir Income Support Programme (BISP). These interventions are largely reactive, responding to dropout after it has already occurred. This study provides a predictive tool capable of flagging high-risk students before they leave school, enabling proactive allocation of limited resources. The two guiding analytic questions are: (1) Which socioeconomic and demographic factors most accurately predict school dropout across Pakistan's provinces? (2) Do the drivers of dropout differ meaningfully between provinces, particularly between Balochistan and the rest?

1.2 Dataset Overview

The data source is UNICEF MICS Round 6, the most recent complete national snapshot for Pakistan, conducted across Punjab (2017-18), Sindh (2018-19), KPK (2019), and Balochistan (2019-20). Two SPSS files are drawn from each province: hl.sav (household members roster, primary source for all education and demographic variables) and hh.sav (household questionnaire, used solely to extract HH15, the interview language variable serving as an ethnicity proxy). The four datasets were merged vertically after confirming column consistency, producing a raw combined dataset of 810,026 individuals. After preprocessing, the final analytical sample comprises 131,748 observations.

2. METHODOLOGY

2.1 Data Preprocessing

Preprocessing followed the seven-step protocol of Khatun et al. (2025) with adaptations for Pakistan-specific SPSS coding. The sample was restricted to ages 6-24 (339,612 rows). Rows with missing or ambiguous target values were dropped (233,736 valid). Placeholder codes were removed, and parental education levels were capped at 4 to eliminate encoding artifacts from provincial SPSS files (364 Mother, 20,680 Father values capped). Mode imputation addressed remaining missing values in Completed_grade (18,826), Last_education_grade (103,151), and parental education fields. After removing 101,687 duplicate records, the final clean sample of 131,748 was retained. ED9 was recoded from 1=YES/2=NO to 1=attending/0=dropout; windex5 was shifted from 1-5 to 0-4.

2.2 Feature Engineering: Age-Grade Mismatch Index

An original feature beyond the Bangladesh paper was engineered: the Age-Grade Mismatch Index (AGM = Age minus (Last_education_grade + 6)). Pakistan's school entry age is 6, so a student's expected age for grade G is G + 6. A 17-year-old in Grade 2 has AGM = +9 (severely behind). A 9-year-old in Grade 3 has AGM = 0 (on track). Values were capped at the legitimate range [-18, 17] to remove 33 encoding artifacts. Last_education_grade was then dropped from the feature set since its information is now captured in AGM, while Age was retained as an independent feature encoding the student's life-stage context. This substitution resolves the perfect linear dependence that would cause VIF = infinity for all three variables simultaneously.

2.3 Feature Selection

A hybrid three-method feature selection approach was applied. VIF confirmed 18 of 20 features fell below the threshold of 5.0; Age and AGM both scored VIF=12.87 due to their correlation, so they were excluded from the VIF-pass set. Logistic regression identified 16 statistically significant predictors ($p<0.05$). Random Forest and XGBoost SHAP analysis each produced a top-10 feature list with strong agreement, with AGM ranking first in both. Under the hybrid voting rule (features appearing in at least 2 of 3 methods are retained), 17 features were selected. Notably, Age and AGM both passed via LR and ML methods despite failing VIF, which is by design: the VIF threshold guards against pure linear redundancy, but moderate correlation between two features that each carry independent signal is acceptable.

2.4 Machine Learning Models

Three models were trained and evaluated: Logistic Regression (LR, scaled via StandardScaler, class_weight='balanced') as the interpretable baseline, Random Forest (RF, n_estimators=100, class_weight='balanced'), and XGBoost (XGB, n_estimators=200, learning_rate=1, max_depth=3, scale_pos_weight=1). All three were evaluated using stratified 10-fold

cross-validation, preserving the 76.4/23.6 class ratio in every fold. For confusion matrix and ROC evaluation, an 80/20 stratified split was applied (105,398 training, 26,350 test).

2.5 Explainable AI: SHAP and LIME

SHAP (SHapley Additive exPlanations) was applied to XGBoost to generate global feature importance summaries and individual-level waterfall explanations for two contrasting student profiles. LIME (Local Interpretable Model-agnostic Explanations) provided independent local explanations for the same profiles. Province-stratified XGBoost models (5-fold CV per province) were trained to compare SHAP feature rankings across Punjab, Sindh, KPK, and Balochistan, directly addressing the second analytic question.

3. RESULTS

3.1 Exploratory Data Analysis

The final analytical sample of 131,748 students reflects a national dropout rate of 23.6%, with 100,719 currently attending (76.4%) and 31,029 classified as dropouts. Figure 1 shows the class distribution. Provincial disparities are the most policy-relevant EDA finding (Figure 2): Balochistan records 32.1% dropout, over 11 percentage points above KPK (21.0%), with Punjab (22.4%) and Sindh (22.8%) near the national average.

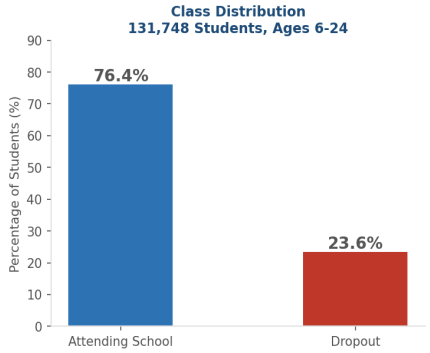

Figure 1: Class Distribution, 131,748 Students

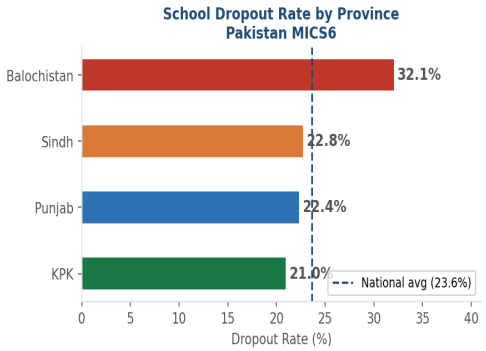

Figure 2: Dropout Rate by Province

Household wealth is a powerful predictor with a steep monotonic gradient (Figure 3): poorest-quintile households reach 38.5% dropout versus 7.3% among the richest. The gender-province heatmap (Figure 4) confirms females face higher dropout in three of four provinces, with the largest gaps in KPK (Female 24.2% vs Male 18.4%) and Balochistan (Female 34.7% vs Male 30.5%). Punjab shows near-equal rates at 22.4% for both sexes.

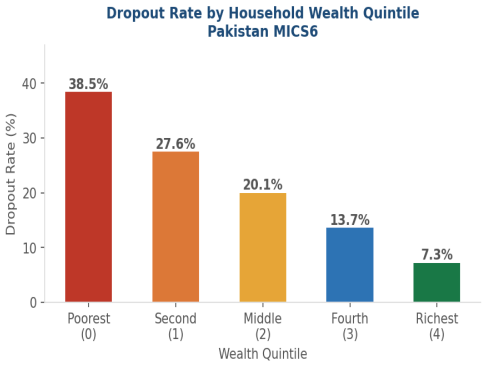

Figure 3: Dropout Rate by Wealth Quintile

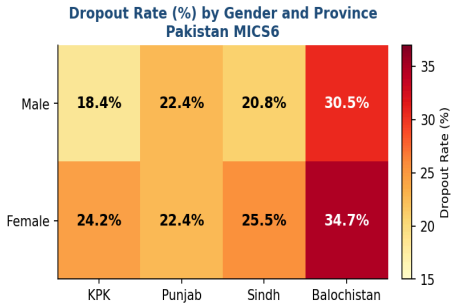

Figure 4: Dropout Rate (%) by Gender and Province

The Age-Grade Mismatch Index reveals a sharply non-linear dropout gradient (Figure 5). Students on track or ahead (AGM <= 0) face only 4.9% dropout. Students severely behind (AGM >= 6) face 44.5% dropout, a 9.1x differential. This is the single most actionable descriptive finding in the analysis: with 55,680 students falling into the severely behind category, a real-time EMIS flag on this metric would identify the highest-risk population at near-zero cost.

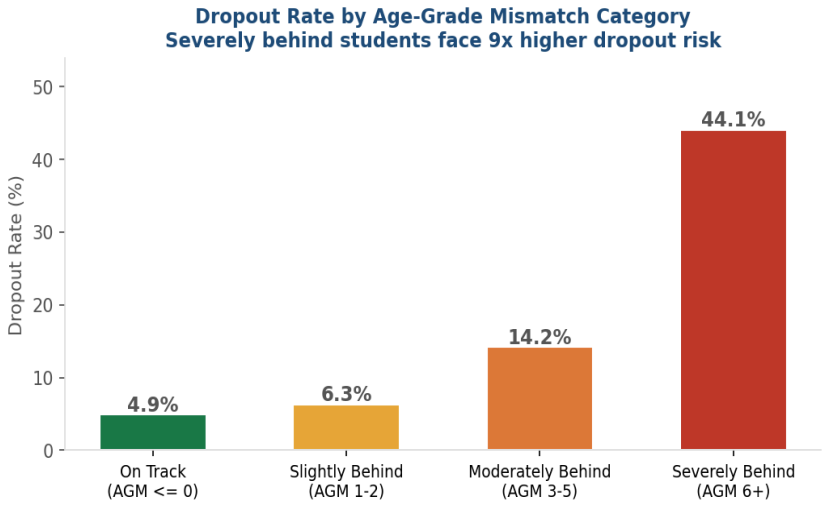

Figure 5: Dropout Rate by Age-Grade Mismatch Category. Severely behind students (AGM >= 6) face a 9.1x higher dropout rate than on-track peers (44.5% vs 4.9%).

3.2 Logistic Regression and Statistical Analysis

Figure 6 presents the odds ratios for key predictors with 95% confidence intervals. Table 1 provides the full statistical results. Completed_grade_YES carries the strongest effect in the model: students who have completed their current grade are approximately 87% less likely to drop out (OR=0.127). The Age-Grade Mismatch Index contributes a per-unit OR of 0.504, meaning each additional year a student falls behind their expected grade multiplies dropout odds by approximately 1.31x (a 23.4% reduction in attending odds per year of mismatch).

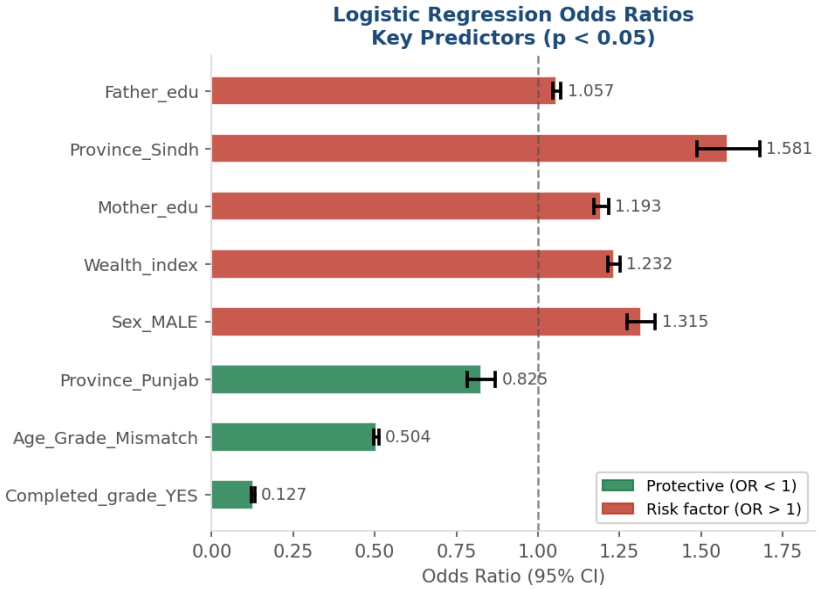

Figure 6: Logistic Regression Odds Ratios with 95% CI. Green bars indicate protective factors (OR < 1); red bars indicate risk factors (OR > 1). All shown predictors are significant at p < 0.0001.

At the provincial level, Sindh carries the highest relative dropout risk (OR=1.581) relative to Balochistan as the reference category, while Punjab is associated with lower relative risk (OR=0.825) net of household controls. Sex_MALE carries OR=1.315, confirming males have higher odds of attending school, meaning females face greater aggregate dropout risk consistent with the raw provincial data.

Table 1: Logistic Regression Results, Key Predictors of School Dropout

Feature	P-Value	Odds Ratio	CI Lower (95%)	CI Upper (95%)
Age	< 0.0001	1.614	1.587	1.642
Completed_grade_YES	< 0.0001	0.127	0.123	0.132
Age_Grade_Mismatch	< 0.0001	0.504	0.496	0.512
Wealth_index	< 0.0001	1.232	1.214	1.251
Mother_s_education	< 0.0001	1.193	1.171	1.216
Sex_MALE	< 0.0001	1.315	1.272	1.360
Father_s_education	< 0.0001	1.057	1.045	1.070
Province_Sindh	< 0.0001	1.581	1.488	1.680
Province_Punjab	< 0.0001	0.825	0.783	0.869
Language_other	< 0.001	0.940	0.907	0.975
Area_URBAN	0.171 *	0.973	0.936	1.012

** Not significant at p < 0.05; retained for domain relevance per feature selection voting.*

3.3 ML Model Performance

Figure 7 presents the ROC curves for all three models on the held-out test set. XGBoost achieves AUC=0.911, outperforming Random Forest (AUC=0.853) and the LR baseline (AUC=0.886). XGBoost achieves 86.7% accuracy, 89.9% precision, 93.1% recall, F1-score of 91.4%, and ROC-AUC of 0.911. The recall of 93.1% is the policy-critical metric: the model correctly identifies 93 out of every 100 actual dropouts, minimizing missed at-risk students when used for targeting. Figure 8 presents the Precision-Recall curves, confirming XGBoost achieves the highest Average Precision (AP=0.965) across all recall thresholds.

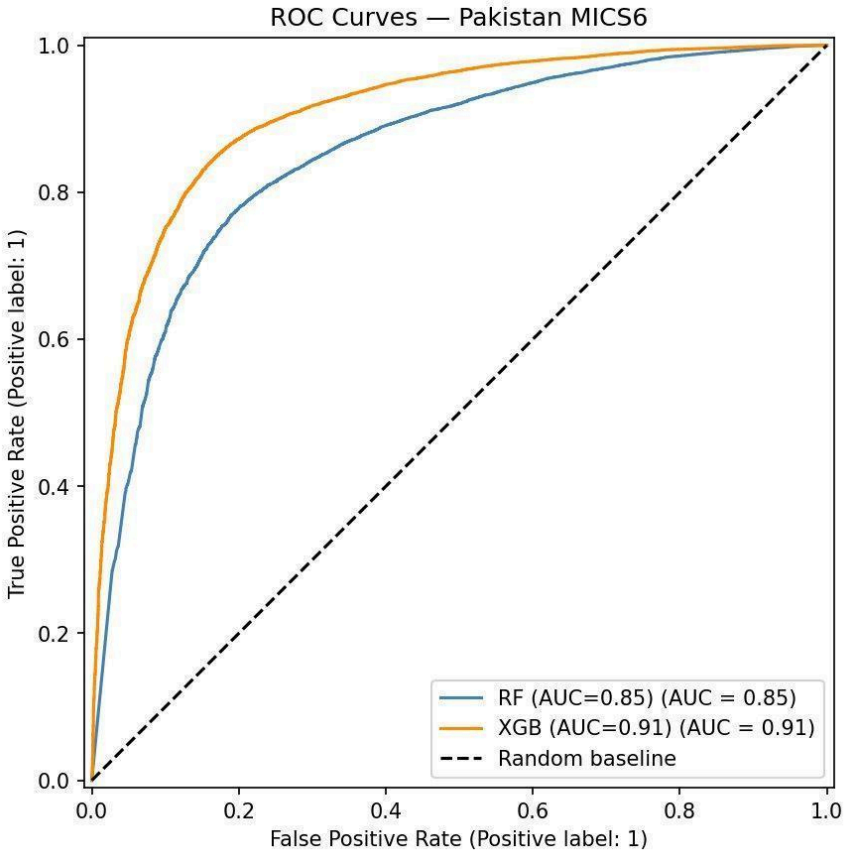

Figure 7: ROC Curves — RF (AUC=0.85) vs XGBoost (AUC=0.91). Both models substantially outperform the random baseline. XGBoost demonstrates superior discrimination across all thresholds.

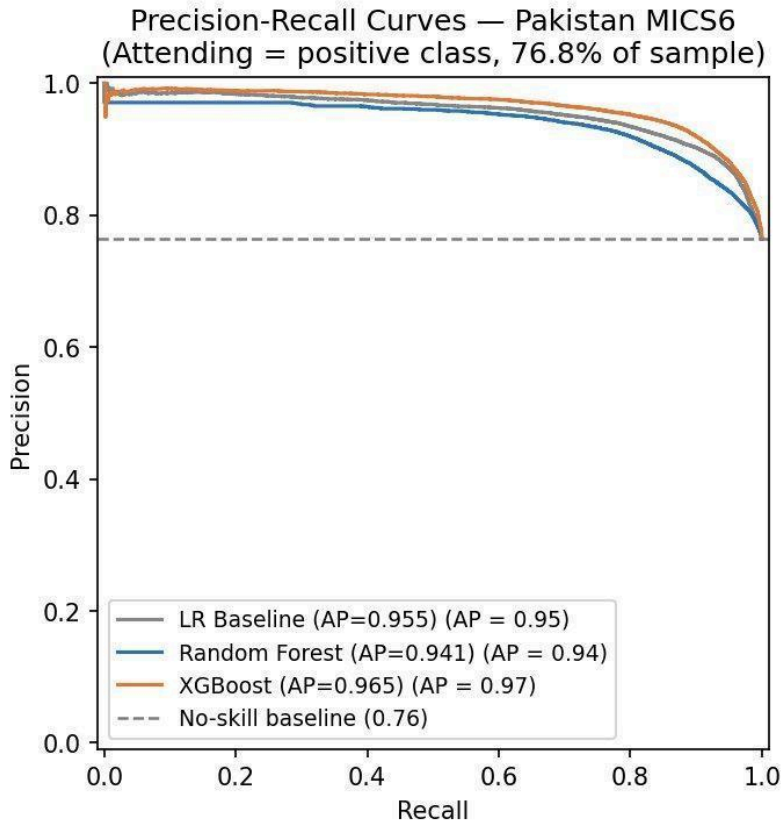

Figure 8: Precision-Recall Curves — LR Baseline (AP=0.955), RF (AP=0.941), XGBoost (AP=0.965). XGBoost achieves the highest Average Precision, maintaining strong precision at high recall levels critical for dropout targeting.

An instructive pattern in the results is that LR (AUC=0.886) outperforms RF (AUC=0.853). This reflects the structure of the feature set: AGM is a near-linear signal (a simple difference of two variables) that logistic regression extracts efficiently, while RF's balanced class weighting over-corrects for the 76/24 imbalance. The AUC gap between LR and XGB is only 0.025, indicating that the dominant relationships in this data are largely linear and XGB's advantage comes from capturing residual nonlinear interactions such as wealth crossed with province.

Table 2: XGBoost 10-Fold Cross-Validation Results

Fold	Accuracy	Precision	Recall	F1-Score	NPV	ROC-AUC
Fold 1	0.8646	0.8987	0.9273	0.9128	0.7370	0.9082
Fold 2	0.8646	0.8975	0.9289	0.9130	0.7397	0.9079
Fold 3	0.8654	0.8939	0.9341	0.9138	0.7501	0.9103
Fold 4	0.8618	0.8993	0.9267	0.9111	0.7324	0.9064
Fold 5	0.8696	0.8989	0.9330	0.9168	0.7532	0.9116
Fold 6	0.8670	0.8963	0.9324	0.9147	0.7490	0.9109
Fold 7	0.8698	0.8998	0.9321	0.9163	0.7517	0.9111
Fold 8	0.8674	0.9004	0.9290	0.9146	0.7434	0.9127
Fold 9	0.8705	0.8987	0.9331	0.9168	0.7545	0.9130
Fold 10	0.8700	0.9017	0.9317	0.9164	0.7513	0.9162
Mean	0.8671	0.8987	0.9308	0.9146	0.7462	0.9109

Table 3: Three-Model Comparison (10-Fold CV Averages)

Model	Accuracy	Precision	Recall	F1-Score	NPV	ROC-AUC
LR (Baseline)	0.8164	0.9291	0.8227	0.8726	0.5805	0.8861
Random Forest	0.8177	0.8851	0.8750	0.8801	0.6090	0.8527
XGBoost (Best)	0.8667	0.8987	0.9308	0.9146	0.7452	0.9107

3.4 Province-Stratified Models

Figure 9 presents the province-stratified XGBoost AUC, directly addressing the second analytic question. The model performs best in Punjab (AUC=0.928) and KPK (AUC=0.912) and worst in Balochistan (AUC=0.840). The 0.088 AUC gap between Punjab and Balochistan is the quantified finding of structural incompleteness: in Balochistan, household survey variables explain less of the dropout variance because key determinants such as school distance, security conditions, and teacher shortages are not captured in MICS questionnaires.

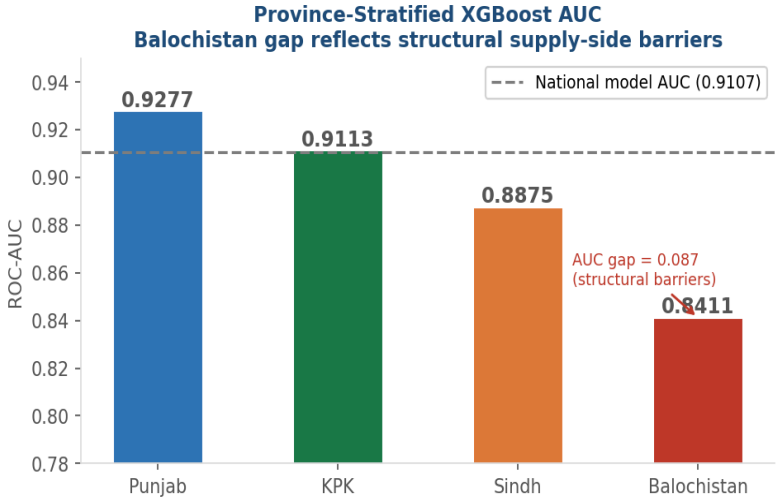

Figure 9: Province-Stratified XGBoost AUC (5-fold CV). The 0.087 gap between Punjab (0.928) and Balochistan (0.841) reflects structural barriers unmeasurable by household surveys.

The province-stratified SHAP rankings reveal one meaningful differentiation across provinces. The top three predictors, Age-Grade Mismatch, Age, and Completed_grade_YES, are consistent across Punjab, Sindh, and KPK, confirming that grade-progression dynamics and age-based attrition are universal mechanisms. The notable departure is Balochistan, where Sex_MALE enters the top five predictors and displaces the parental education variables that rank in the top five elsewhere. This indicates that in Balochistan, gender operates as an independent household-level constraint on attendance beyond what wealth and academic progression alone can explain, consistent with the raw data showing Balochistan's gender gap (Female 34.7%, Male 30.5%) as the widest of any province. In Punjab, Sindh, and KPK, Wealth_index and Mother_s_education rank fourth and fifth respectively, suggesting that once grade progression is controlled for, economic and maternal education factors drive the residual variation similarly across these three provinces.

3.5 SHAP and LIME Interpretability

Figure 10 presents global SHAP and RF feature importance for the national model. AGM ranks first in both methods (SHAP mean=2.501, RF importance=0.265), followed by Age (SHAP=1.462) and Completed_grade_YES (SHAP=0.868). The strong agreement validates the hybrid feature selection approach.

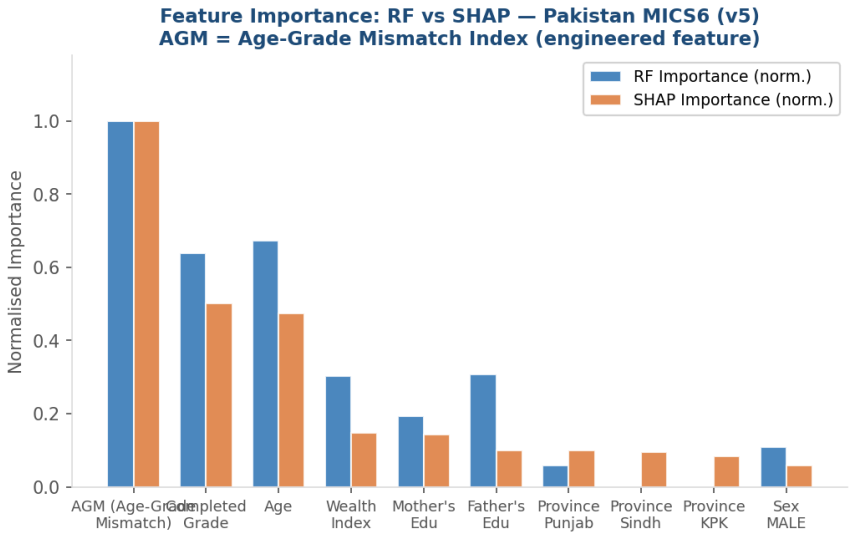

Figure 10: Normalised Feature Importance across Random Forest and XGBoost SHAP. AGM = Age-Grade Mismatch Index (engineered feature). Both methods agree on the top three predictors.

At the individual level, the SHAP waterfall for the most confident attending-student prediction depicts a 6-year-old male in KPK (AGM=-3, ahead of expected grade), richest wealth quintile, with highly educated parents and a non-majority language background. Stacked protective factors drive a 100% attendance prediction. The highest-confidence dropout profile depicts a 17-year-old female in Punjab (AGM=10, severely behind expected grade), poorest wealth quintile, with no parental education. The AGM of 10 means this student is 10 years older than expected for their completed grade, a severe age-grade mismatch consistent with the pattern of delayed entry compounded by grade repetition. The LIME analysis corroborates both profiles, with AGM and Completed_grade_YES consistently ranking as the top local contributors in both directions.

4. POLICY IMPLICATIONS

The empirical findings generate four recommendations, each grounded in a specific model output, linked to an implementable government action, and paired with a measurable evaluation indicator.

- **Age-grade mismatch flagging in NEMIS.** The Age-Grade Mismatch Index is the strongest predictor of dropout across all methods, with severely behind students facing a dropout rate 9.1 times higher than on-track peers (44.5% versus 4.9%). The National Education Management Information System, administered by the Ministry of Federal Education and Professional Training, already captures enrolled age and current grade at the school level. Adding AGM as a computed field requires no new data collection, only a software update to the NEMIS architecture and a standardized counselor referral protocol for flagged students. Lewin (2009) and the World Bank (2018) both document over-age enrollment as the strongest proximate dropout predictor in South Asian systems, and note that schools rarely intervene because no institutional mechanism exists to surface the signal. This recommendation operationalizes that mechanism. Evaluation indicators: share of AGM-flagged students receiving documented counselor contact within 30 days; dropout rate in flagged versus matched unflagged cohorts after one academic year.

- **Risk-adjusted targeting for BISP Taleemi Wazaif.** The model shows that within the poorest wealth quintiles, dropout risk varies substantially by AGM. A two-dimensional eligibility screen, combining the existing National Socioeconomic Registry poverty threshold with an AGM score drawn from NEMIS enrollment records, would redirect existing transfer capacity toward the highest-risk children within the already-eligible poverty pool. Matching is feasible through child B-form CNIC identifiers, which both NEMIS and the National Socioeconomic Registry already collect. A Memorandum of Understanding between MoFEPT and the BISP Secretariat, within existing executive authority, constitutes the full regulatory requirement. Behrman, Parker, and Todd (2011) document that conditional cash transfers achieve their highest schooling impact when directed at academically delayed students facing labor market opportunity costs, precisely the population AGM identifies. Evaluation indicator: 12-month school retention rate for AGM-targeted beneficiaries relative to income-eligible, non-targeted controls.

- **Supply-side investment in Balochistan.** Balochistan records both the highest dropout rate (32.1%) and the lowest province-stratified model AUC (0.841), a gap of 0.087 relative to Punjab. The reduced model accuracy is

direct evidence that unmeasured structural barriers, including school distance, teacher absence, and security conditions, drive Balochistan dropout beyond what household survey variables can explain. Andrabi, Das, and Khwaja (2013) document that each additional kilometer to school reduces female enrollment by 1.5 percentage points in rural Pakistan, and ASER Pakistan (2023) records that 40% of rural Balochistan settlements lack a functional government school within 3 kilometers. Demand-side transfers cannot resolve a supply-side absence. The Balochistan Elementary and Secondary Education Department, with capital financing through the federal Public Sector Development Programme and co-financing from the active World Bank Balochistan Human Capital Investment Project, should prioritize school construction in unserved union councils and mandatory female teacher recruitment quotas for rural postings. Evaluation indicator: share of girls residing within 3 kilometers of a functional school; provincial dropout rate at 3-year intervals aligned with MICS survey rounds.

- **Father engagement for girls' secondary retention in KPK.** KPK records the largest gender gap of any province (Female 24.2%, Male 18.4%). The province-stratified SHAP analysis shows that the gender gap in KPK operates through grade-progression mechanisms rather than as a direct independent predictor at the household level. The raw gap itself, 5.8 percentage points between female and male dropout, is the largest of any province and is consistent with patriarchal household decision-making shaping whether girls continue to secondary school, a dynamic supported by community-level evidence in the literature. Hasan (2010) documents an 18% increase in girls' secondary enrollment in KPK communities following structured father engagement campaigns led by community and religious leaders. The KPK Elementary and Secondary Education Department should expand the existing School Management Council framework to include quarterly paternal engagement sessions for households with girls enrolled in Grades 6 through 10, targeting father attitudes as the mediating behavioral variable. This is an addendum to the operational KP Education Sector Plan rather than a new program structure. Evaluation indicator: female secondary enrollment rate and KPK gender gap in dropout at 2-year intervals; SMC father participation rate in program schools.

5. LIMITATIONS AND ETHICAL CONSIDERATIONS

Several limitations constrain interpretation. First, the data is cross-sectional: the model identifies correlates of dropout, not causal mechanisms. Longitudinal tracking would be required to build a true prospective early-warning system. Second, three variables had substantial missing rates (Last_education_grade 44%, parental education 29% each) imputed by mode, likely understating educational disadvantage among the most at-risk children. KNN imputation would better preserve relationship structures and is recommended for future iterations. Third, UNICEF MICS survey weights were not applied, consistent with the Bangladesh paper's approach for predictive modeling, so point estimates are not population-representative. Fourth, the education level capping (20,680 Father values capped to 4) corrected a data artifact but also standardized a small proportion of records.

Regarding fairness and ethics: the model systematically assigns higher dropout risk to students from Balochistan, rural areas, and minority language groups. These patterns reflect real structural inequalities, but deploying model outputs in resource allocation must remain additive, directing additional support toward high-risk students rather than reducing services. Using dropout risk scores to deprioritize any province or demographic group would entrench existing disadvantage. All MICS data used is fully anonymized; no identifiable participant information was accessed at any stage.

6. CONCLUSION

This study demonstrates that a hybrid machine learning and explainable AI framework can accurately predict school dropout risk in Pakistan using nationally representative MICS Round 6 data. XGBoost achieves 86.7% accuracy and a ROC-AUC of 0.911 across 131,748 school-age children from four provinces. The engineered Age-Grade Mismatch Index is the single most powerful predictor, with students severely behind their expected grade facing dropout rates 9.1x higher than on-track peers. The province-stratified analysis quantifies what policymakers have long suspected: Balochistan requires fundamentally different interventions from Punjab, and the model's AUC gap (0.928 vs 0.841) provides a statistical basis for that distinction. Combined with the gender and wealth findings, this analysis provides a multi-dimensional evidence base for BISP targeting, EMIS early-warning system design, and provincial education budget prioritization.

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